2.3. Representations

Definition 2.3.1. A **representation** of an algebra A (also called a **left A-module**) is a vector space V together with a homomorphism of algebras $\rho : A \to \mathrm{End}V$.

Similarly, a **right A-module** is a space V equipped with an antihomomorphism $\rho : A \to \mathrm{End}V$; i.e., ρ satisfies $\rho(ab) = \rho(b)\rho(a)$ and $\rho(1) = 1$.

The usual abbreviated notation for $\rho(a)v$ is av for a left module and va for a right module. Then the property that ρ is an (anti)homomorphism can be written as a kind of associativity law: $(ab)v = a(bv)$ for left modules, and $(va)b = v(ab)$ for right modules.

Remark 2.3.2. Let M be a left module over a commutative ring A. Then one can regard M as a right A-module, with $ma := am$. Similarly, any right A-module can be regarded as a left A-module. For this reason, for commutative rings one does not distinguish between left and right A-modules and just calls them A-modules.

Here are some examples of representations.

Example 2.3.3. 1. $V = 0$.

2. $V = A$, and $\rho : A \to \mathrm{End}A$ is defined as follows: $\rho(a)$ is the operator of left multiplication by a, so that $\rho(a)b = ab$ (the usual product). This representation is called the **regular** representation of A. Similarly, one can equip A with a structure of a right A-module by setting $\rho(a)b := ba$.

3. $A = k$. Then a representation of A is simply a vector space over k.

4. $A = k\langle x_1, \ldots, x_n\rangle$. Then a representation of A is just a vector space V over k with a collection of arbitrary linear operators $\rho(x_1), \ldots, \rho(x_n) : V \to V$ (explain why!).

Definition 2.3.4. A **subrepresentation** of a representation V of an algebra A is a subspace $W \subset V$ which is invariant under all the operators $\rho(a) : V \to V$, $a \in A$.

For instance, 0 and V are always subrepresentations.